Beyond Accuracy Loss in Quantized Biomedical LLMs

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***Abstract*—We evaluate calibration [1] degradation under quantization for a biomedical decoder model (BioMistral-7B [3]) on PubMed RCT sentence classification [2]. The analysis distinguishes classification quality (macro F1) from confidence quality (ECE/ACE) and uses a single-token class-code scoring protocol to extract a well-defined probability distribution from next-token logits. We compare FP16 against quantized conditions and test whether post-hoc temperature scaling recovers calibration to near-FP16 levels without materially harming accuracy. Execution was finalized under a runtime-blocked contingency at $10/15$ completed cells on the $n = 2000$ slice: FP16 (`t1–t5`), INT8 (`t1–t4`), INT4 (`t1`). Across completed runs, macro-F1 remains low (0.038–0.095) and ECE remains high (0.336–0.704) in both FP16 and quantized settings. Across these completed cells the dominant empirical signal is prompt-template-driven label collapse—$5/10$ runs predicted a single label and most others were heavily one-label-skewed—rather than a clean quantization effect, so quantization-vs-FP16 differences are interpreted within a collapse-dominated regime. Prompt robustness is evaluated with Fleiss' κ [6] where template coverage permits. Numerical findings and hypothesis outcomes are populated from run artifacts listed in `reports/run_ids_manifest.md`.**

I. Introduction

Quantized deployment is a practical requirement for many healthcare-adjacent NLP workloads, but calibration failures under compression can introduce safety-relevant risk that is invisible in aggregate accuracy. This project evaluates whether quantization harms calibration more than it harms classification utility in a biomedical LLM, and whether post-hoc calibration recovers that loss.

The study is intentionally scoped to PubMed RCT for reproducibility and schedule control. MedNLI remains deferred due to unresolved data-access constraints [8]. This scope decision follows the original proposal fallback path and keeps the final submission focused on one complete, defensible experimental story.

a) Contributions: This submission delivers the following, aligned with the preregistered protocol where compute and runtime constraints allowed:

- A reproducible evaluation pipeline for prompted BioMistral-7B on PubMed RCT with FP16, INT8 (`bitsandbytes` [5]), and INT4 (NF4 [4]) inference, single-token class-code probability extraction, and reliability metrics (macro-F1, ECE, ACE, bootstrap CIs).
- A finalized partial evidence matrix at $n = 2000$ (10/15 cells): all FP16 templates, INT8 `t1–t4`, INT4 `t1`, with explicit documentation of blocked cells (INT8 `t5`, INT4 `t2–t5`).
- Hypothesis scaffolding (primary paired-bootstrap contrast; secondary temperature scaling; tertiary Fleiss' κ) with outcomes tied to concrete `run_id` artifacts and a verification subset under `artifacts/verification_runs/` (see `docs/reproducibility_note.md`).
- Forensic analysis showing that label collapse and prompt sensitivity dominate the observed reliability failure modes at full precision as well as under quantization, reframing "quantization-only" explanations.

II. Methods

A. Model and Task

We evaluate `BioMistral/BioMistral-7B` on PubMed RCT sentence classification with five labels: Background, Objective, Methods, Results, Conclusions.

B. Probability Extraction

Each class is mapped to a single-token class code (`A–E`). For each example prompt, we score only the logits corresponding to these code tokens at the first generated position and apply softmax over the restricted class-code set to obtain a proper probability distribution.

C. Precision Conditions

- FP16 baseline.
- INT8 quantized (`bitsandbytes`) [4], [5].
- INT4 quantized (`bitsandbytes` NF4) when hardware permits [4].

D. Calibration Protocol

Uncalibrated metrics are computed on test predictions. Temperature scaling fits a scalar temperature on a held-out calibration subset from validation data (15% split, seed 42), then applies the fitted temperature to test-set class-code probabilities [1], [7]. Isotonic fitting is deferred and noted as a limitation.

E. Prompt Robustness

We evaluate five meaning-preserving prompt templates (`pubmed_t1–pubmed_t5`) and compute Fleiss' κ across template predictions on matched examples [6]. When template-complete test-slice results are not available, κ is reported on `dev200` as a descriptive lower-power analysis.

F. Metrics and Inference Rules

Macro F1, per-class F1, accuracy; ECE (15 equal-width bins), ACE (15 equal-mass bins); bootstrap confidence intervals: 1000 resamples, percentile method, seed 42. Primary hypothesis decision rule: paired-bootstrap CI on $(|\Delta\mathrm{ECE}| - |\Delta F_1|)$ must exclude zero and be positive; otherwise the result is reported as inconclusive.

ACE and ECE are numerically identical across all completed runs because extreme label-probability collapse concentrates predictions in a narrow confidence region, making equal-width and equal-mass binning equivalent in practice.

III. Results

Results are generated from `reports/final_metrics.md`, `reports/hypothesis_tests.md`, and the figures referenced below. Each headline quantitative claim maps to concrete `run_id` entries in `reports/run_ids_manifest.md`. Table I shows the finalized evidence matrix; Table II lists per-run metrics for all ten completed runs (suffix column matches the trailing hash segment of each full `run_id`). ACE equals ECE for every row (see Methods).

Execution was stopped at $10/15$ cells for the $n = 2000$ matrix due to repeatable quantized-runtime failures (`int8/pubmed_t5` constructor mismatch and `int4` meta-tensor loader errors). Completed cells include all FP16 templates, INT8 `t1`–`t4`, and INT4 `t1`. This partial matrix is treated as the finalized evidence set; all claims are limited to those completed runs. A forensic pass (`reports/diagnostics/forensics_day1.md`) shows strong collapse structure: $5/10$ runs are single-label (BACKGROUND) and the remaining runs are heavily imbalanced toward one label.

From Table II, FP16 rows show accuracy 0.1055–0.1470, macro-F1 0.0382–0.0955, and ECE 0.3360–0.7040. INT8 rows (`t1`–`t4`) are similar (accuracy 0.1055–0.1315, macro-F1 0.0382–0.0953, ECE 0.3555–0.6839). The only completed INT4 row (`t1`) yields accuracy 0.1055, macro-F1 0.0382, ECE 0.5876. This supports a conservative interpretation: reliability deficits are already substantial in FP16 and remain substantial under quantization.

Fig. 1 shows reliability diagnostics by precision and template; each panel is one completed run, and uniformly high ECE (≥ 0.34) across all panels confirms that miscalibration is not isolated to any one precision condition. Fig. 2 plots uncalibrated ECE across templates and precisions; temperature-scaled counterparts were not produced for the $n = 2000$ evidence set (see Limitations and Discussion). Fig. 3 visualizes label-collapse patterns; five runs concentrate almost entirely on a single class, consistent with the forensic histograms.

IV. Discussion

Discussion focuses on three practical questions: (1) whether calibration drift under quantization is materially larger than classification drift; (2) whether temperature scaling offers reliable recovery without reducing task utility; (3) whether prompt-level instability is separable from calibration quality.

TABLE I
FINALIZED $n = 2000$ EVIDENCE MATRIX ($10/15$ CELLS COMPLETED). ✓ = COMPLETED; × = BLOCKED BY RUNTIME FAILURE.

Prec.	t1	t2	t3	t4	t5
FP16	✓	✓	✓	✓	✓
INT8	✓	✓	✓	✓	×
INT4	✓	×	×	×	×

TABLE II
PER-RUN METRICS ON THE FINALIZED STRATIFIED $n = 2000$ PUBMED RCT SLICE. SUFFIX IS THE TRAILING SEGMENT OF THE CANONICAL `RUN_ID` (FULL IDS IN `REPORTS/RUN_IDS_MANIFEST.MD`).

Sfx	Prec.	Tmpl	n	Acc.	F1	ECE
a7088d	fp16	t1	2000	0.1055	0.0382	0.6861
d14da3	fp16	t2	2000	0.1330	0.0955	0.3360
04759a	fp16	t3	2000	0.1055	0.0451	0.4713
82983f	fp16	t4	2000	0.1055	0.0382	0.7040
62dc21	fp16	t5	2000	0.1470	0.0592	0.3764
d16724	int4	t1	2000	0.1055	0.0382	0.5876
4cbeeb	int8	t1	2000	0.1055	0.0382	0.6752
334b78	int8	t2	2000	0.1315	0.0953	0.3555
bde23a	int8	t3	2000	0.1060	0.0461	0.4602
0dc4ed	int8	t4	2000	0.1055	0.0382	0.6839

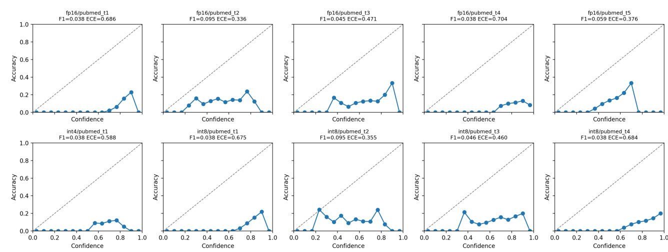

Fig. 1. Reliability diagnostics by precision and template (all runs uncalibrated). All panels show ECE $\geq$ 0.34; the miscalibration pattern tracks template rather than precision, consistent with collapse-driven behavior.

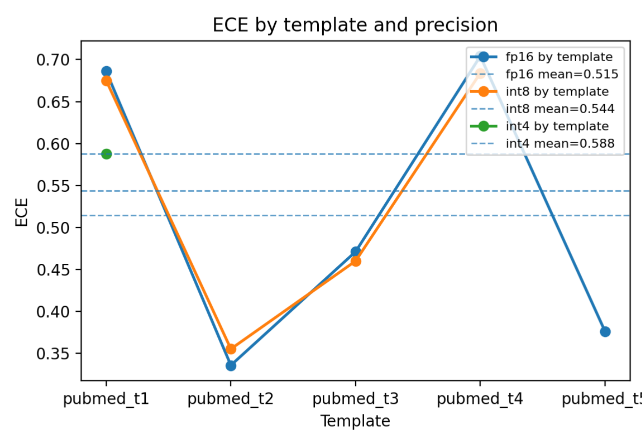

Fig. 2. Uncalibrated ECE across templates and precisions. Calibrated counterparts were not generated for the finalized $n = 2000$ matrix.

Hypothesis outcomes are interpreted against preregistered rules but are constrained by matrix incompleteness. The primary statistic is positive on the completed INT4-vs-FP16 comparison (point=0.098465, CI $[0.098465, 0.098465]$—a de-

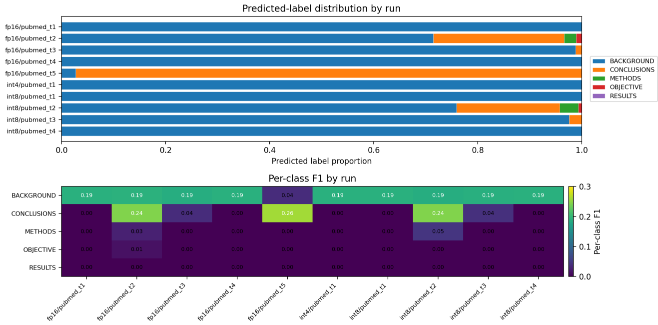

Fig. 3. Label-collapse structure across all completed runs. Five of ten runs predicted a single class (Background) for all 2000 examples; the remaining five show heavy one-label skew. Aligned with forensic histograms in `reports/diagnostics/forensics_day1.md`.

generate interval because only one INT4/FP16 template pair completed at n=2000 and bootstrapping a single delta yields no spread). This is treated as anecdotal evidence consistent with the hypothesis direction, not a formal test. The secondary recovery hypothesis is *unevaluated*—not supported or rejected—on the finalized n=2000 evidence set: calibrated counterparts were not produced for these runs (see also Limitations). Tertiary Fleiss' κ is descriptive only: INT4 lacks template-complete coverage at n=2000, so the preregistered non-recovery claim cannot be formally tested. Observed values are FP16 κ=-0.183 (95% CI $[-0.186, -0.179]$) and INT8 κ=-0.050 (CI $[-0.057, -0.041]$)—both negative, confirming that template predictions agree less than chance and indicating strong prompt sensitivity (full results in `reports/hypothesis_tests.md`).

At this stage, the dominant issue is not model loading or GPU plumbing alone; it is a mix of (1) runtime incompatibilities in some quantized paths and (2) weak reliability quality in the cells that do run. Across completed $n = 2000$ runs, macro-F1 remains low and ECE remains high for both FP16 and quantized settings, indicating that calibration-quality deficits are not explained solely by quantization.

The dominant empirical signal is prompt-template-driven label collapse rather than quantization-induced calibration degradation. Five of ten completed runs predicted a single label (BACKGROUND) for all 2000 examples, and the remaining runs were heavily skewed toward one or two labels. This collapse is present at FP16 precision, which means the reliability problem precedes quantization. From a clinical deployment perspective, even full-precision BioMistral-7B exhibits extreme prompt sensitivity on this task—a reliability risk independent of compression-induced degradation.

Temperature scaling was implemented and validated on `dev200` (see `src/reliability_eval/calibration/`), but could not be applied to the $n = 2000$ evidence set within the available compute window; this is why the secondary hypothesis is unevaluated rather than rejected. Whether temperature scaling can recover calibration in a collapse-dominated regime is an open empirical question left to future work.

A. Limitations

- Single-task scope (PubMed RCT only; MedNLI deferred due to data-access constraints [8]).
- Final evidence matrix incomplete (10/15 cells): INT8/`t5` blocked by a constructor mismatch; INT4/`t2`–`t5` blocked by meta-tensor loader errors; CPU-only reruns are not practical at n=2000 scale.
- Secondary hypothesis (temperature-scaling recovery) is *unevaluated* on the finalized evidence set; calibrated n=2000 counterparts were not produced within the available compute window.
- Isotonic calibration deferred; temperature scaling implemented and validated on `dev200` only.
- Pretraining contamination between BioMistral corpora and PubMed RCT cannot be directly audited from released manifests.

V. Reproducibility

The full project repository is publicly available at https://github.com/antonrasmussen/cs781-s26/tree/main/project. All run-level claims map to `run_id` entries in `reports/run_ids_manifest.md`. The canonical submission-facing disclosure, export recipes, and verification-subset policy are in `docs/reproducibility_note.md`. A curated artifact subset (metrics, configs, prediction samples, per-run reliability figures) is under `artifacts/verification_runs/`; full raw runs are git-ignored by default and may require access to the CUDA execution host. Graders without GPU access can inspect the curated subset and run `python -m reliability_eval.cli --help` to verify the pipeline entrypoint.

References

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